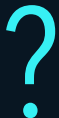


Can a search agent earn robustness under hidden loss?

A controlled agent environment for compact linear-optical circuit discovery, where every query, budget decision and stopping rule can be replayed.



RESEARCH QUESTION

Can an agent discover circuits that retain useful fidelity and success probability under unseen loss patterns - while obeying fixed component and simulation budgets?

ENVIRONMENT OBJECT Circuit topology, phase settings and non-ideal optical loss	SIMULATOR Perceval 1.2.4	PAIRING 12 seeded episodes	EQUAL COST 48 queries per policy
--	--	--	--

EXECUTED RESULT
DISCOVERY
GATE FAILED

0.8574

Random mean held-out worst-quartile fidelity

0.8673

Greedy mean held-out worst-quartile fidelity

0.7971

Budget-Aware Diversity Search mean; no advantage claimed

CONTROLLED RUN

Query trace, not a leaderboard.

RESET frozen target + budget + perturbation schedule

POOL 192 valid candidates / episode

WARMUP 6 shared queries

TRAIN 16 loss patterns

FREEZE chosen circuit + Pareto archive

HOLDOUT 24 stronger shifted patterns

TRACE action, reason, reply, time, budget

INTEGRITY BOUNDARY

Hidden labels remain outside the agent. A controller validates actions and writes a complete JSONL audit trail. The MIT harness, dependency lock, raw traces, figures, tables and checksums are retained; one command reruns the policies and QA. Policy modules can be swapped without changing the environment.

AGENT WORKFLOW

A routed experiment loop

Q0 / 02

OBSERVE – PROPOSE – SIMULATE – UPDATE – AUDIT
fixed seed / fixed query budget / frozen gate

01 / STEWARD

Freeze the world

Load the split, allowed catalogue, budget and perturbation schedule.

02 / CANDIDATE

Choose one query

Add or edit a valid component and state the search objective it serves.

03 / ORACLE

Run permitted physics

Return fidelity, success probability, depth and any constraint violation.

04 / EVIDENCE

Update the belief

Recompute trajectory, uncertainty, diversity and remaining-resource summaries.

06 / AUDITOR

Apply the gate once

Evaluate hidden loss patterns after selection; retain wins and failures alike.

05 / STOP

Freeze the candidate

End only when the budget is exhausted or the registered stopping rule fires.

PREDECLARED DISCOVERY SIGNAL

A non-dominated circuit must improve held-out worst-quartile fidelity or survival over random and greedy search at equal cost, while respecting the component budget and retaining rank under shifted loss.

FIDELITY CONDITION

+0.010

Mean practical margin over each baseline, with paired 95% CI lower bound above zero.

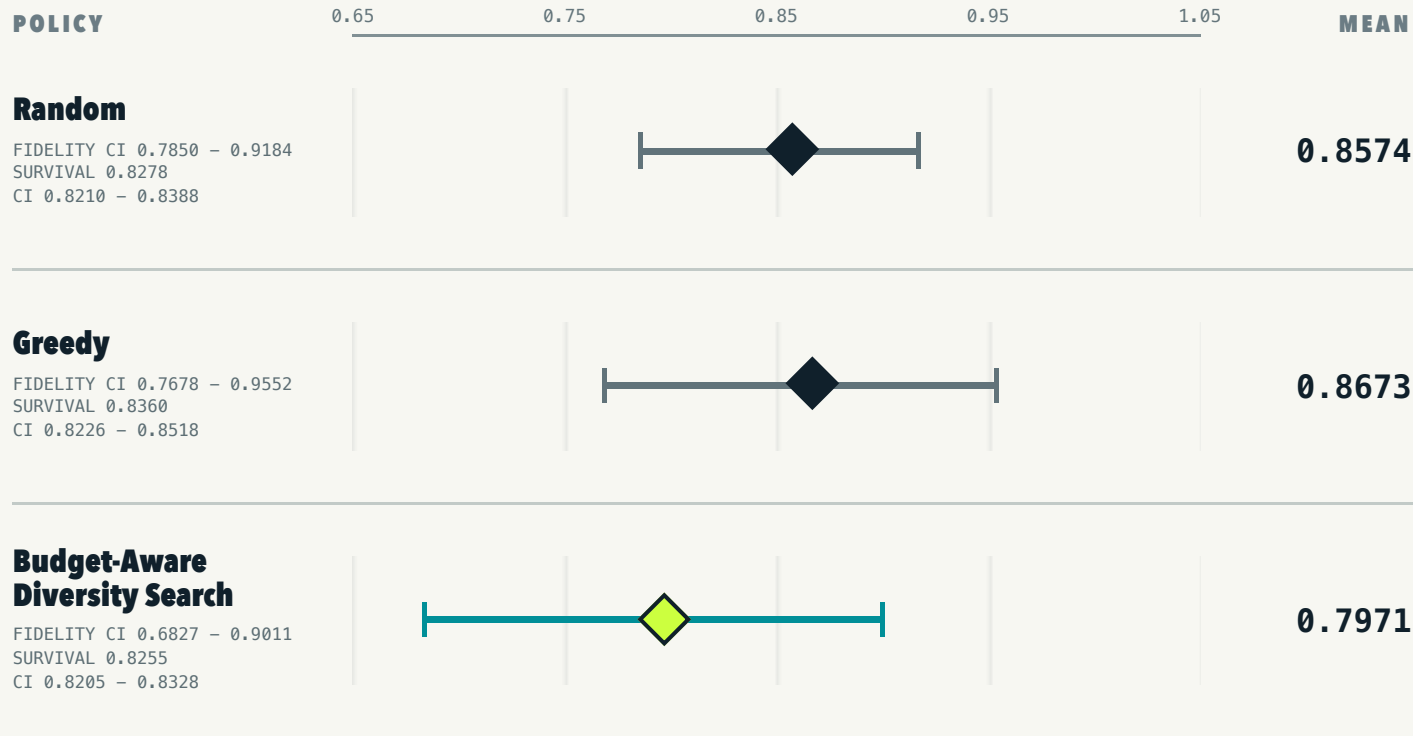
SURVIVAL CONDITION

-0.010

Paired 95% CI lower bound must remain above the non-inferiority margin.

EXECUTED EVIDENCE / POST-FREEZE EVALUATION

The search policy did not beat simpler baselines.

HELD-OUT WORST-QUARTILE FIDELITY
12 PAIRED EPISODES / MEAN + 95% CI

BADS MINUS RANDOM / FIDELITY

-0.0602

Paired 95% CI: -0.1659 to 0.0383. BADS won 6 of 12 episodes, with one tie. Survival difference: -0.0024 [-0.0152, 0.0081].

BADS MINUS GREEDY / FIDELITY

-0.0702

Paired 95% CI: -0.1366 to -0.0159. BADS won 0 of 12 episodes, with six ties. Survival difference: -0.0106 [-0.0262, 0.0000].

Boundary. The deterministic single-photon transfer-matrix model excludes partial distinguishability, multiphoton interference, detector response, source impurity, fabrication tolerance and hardware calibration. The finite pool centres on one disclosed target; 12 episodes estimate uncertainty but remain a small sample.

DISCOVERY AUDITOR

NOT PASSED

The acquisition rule did not show a reliable, practically sized held-out fidelity advantage over both baselines. The negative result is retained without removing unfavourable episodes or changing the gate.

NEXT REGISTERED EXPERIMENT

- Separate uncertainty-only and diversity-only ablations.
- Add multiple target families and a topology-family holdout.
- Introduce source, detector and partial-distinguishability noise.
- Keep the gate fixed; reserve hardware claims for calibrated devices.

Perceval 1.2 documentation:

perceval.quandela.net/docs/v1.2/

Source and licence: github.com/Quandela/Perceval

Exact simulator release: pypi.org/project/perceval-quandela/1.2.4/